Predicting Bilingual Proficiency: Eye-Tracking Analysis of Chinese-English Code-Switching Using Machine Learning

Selina Hui, Lily Wu, Asha Dukkipati, Sienna Jackson

Abstract

This study explores how machine learning (ML) analysis of eye-tracking data can reveal the relationship between code-switching and visual attention in bilinguals. Code-switching, the switching of languages during communication, offers insight into the cognitive mechanisms underlying bilingual processing. By analyzing how bilinguals process code-switched Chinese-English sentences, our research uses a Random Forests classification model to predict second language proficiency based on eye movement. This approach combines eye-tracking measurements with pattern recognition to identify how attention patterns during language switching correlate with proficiency levels. Our interdisciplinary methodology merges cognitive science, linguistics, and computational modeling to reshape how we understand bilingual language control. Predicting proficiency using real-time data enables more efficient language assessment tools for educational and clinical settings. Insights into cognitive load during language-switching can inform personalized teaching strategies, while this research also advances AI systems for bilingual language use, improving machine translation, natural language processing, and human-computer interactions. By examining the relationship between visual attention and language proficiency, the study aims to uncover how humans navigate multiple language systems and has far-reaching implications in an increasingly multilingual world.